BTS-I&II(R/S)-(04/09)-21-0806 Reg. No.

C

B.Tech. Degree I&II Semester Regular/ Supplementary Examination April/ September 2021

19-200-0103A & 19-200-0203B ENGINEERING GRAPHICS

(2019 Scheme)

(Regular/ supplementary for EC and supplementary for other branches)

Time: 3 Hours Maximum Marks: 60

PART A

($1 \times 12 = 12$)

I. Construct a backward Vernier scale of RF = 1/50000 showing kilometres, hectometres and decametres and long enough to measure up to 5km. Mark distance of 2.79 km and 3.56 km.

OR

II. The foci of an ellipse are 100 mm apart and the minor axis is 80 mm long. Measure the length of the major axis. Draw the ellipse by using intersecting arc method. Also draw a tangent to the ellipse from any point outside the ellipse.

PART B

($4 \times 15 = 60$)

III. A line AB 80 mm long has a length of 60 mm in top view and 70 mm in front view. If one end is 15 mm above HP and 10 mm in front of VP. Determine the inclinations of the line with HP and VP. Also mark its traces.

OR

IV. A rectangular plate 65 mm × 45 mm has one of its shorter edges in the VP and inclined at 40° to HP. Draw its top view if its front view is a square of side 45 mm.

V. A pentagonal prism base edge 25 mm height 50 mm resting on one of its corner on HP with the axis inclined at 30° with HP and 40° with VP. The nearest point of the solid is at a distance of 15 mm away from VP. Draw its projections.

OR

VI. A cone of base diameter 40 mm and axis 55 mm is resting on its base on HP. It is cut by a section plane inclined at 40° to VP and passing through a point 10 mm in front of the axis. Draw the front view, top view and true shape of the section.

OR

VII. Draw the development of the lateral surface of a right regular hexagonal pyramid of 40 mm side and height 80 mm. It stands on HP with one of its base edge perpendicular to VP. A circular hole of 35 mm diameter is drilled through the pyramid in such a way that the axis of the hole is perpendicular to VP and 20 mm above the base. Assume that the axes intersect each other.

OR

VIII. A vertical cylinder of diameter 125 mm and 250 mm long is completely penetrated by a horizontal cylinder of diameter 100 mm and 230 mm long. The axis of the horizontal cylinder is parallel to VP and is 7 mm in front of the axis of the vertical cylinder. Draw the projections of solid showing line of intersection.

IX. A hemisphere of 40 mm diameter is nailed on the top of a hexagonal prism with flat surface up. Base edge of the prism is 30 mm and height is 60 mm. The axes of both solids coincide. Draw the isometric projection.

OR

X. A square pyramid 35 mm side and 60 mm height is resting on the ground with its base, with base edges parallel to picture plane and 10 mm behind it. The station point is 50 mm in front of picture plane and 80 mm above ground plane. The central plane is 40 mm towards the right of the centre of the pyramid. Draw its perspective projection.

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